

Title:**A Real-Time Haptic Simulation Prototype for Liver Lesion Localization Training: System Design, Technical Feasibility Validation, and Formative Expert Evaluation**

The authors present a relevant and practically meaningful prototype for real-time haptic simulation in liver lesion localization training. The manuscript is generally well organized, and the system design clearly explains how GB-cFEM, bimanual tracking, and a fingertip-worn haptic interface are integrated into a closed-loop training environment. The technical validation and formative expert feedback provide useful early evidence of system feasibility. However, the current version still requires substantial improvement before publication. In particular, the novelty over the authors' prior work needs to be clarified, the haptic output requires stronger quantitative validation, and the evaluation with only one expert is not sufficient to support broader training or clinical-use claims. Overall, the work has merit, but further updates are needed to strengthen the technical evidence, evaluation depth, and alignment between claims and results.

Strengths

1. The paper addresses a relevant and practical problem in medical simulation, especially for liver lesion localization where tactile feedback is important.
2. The system architecture is clearly presented and connects the physics simulation, hand tracking, haptic rendering, and task setup in a logical way.
3. The integration of GB-cFEM with a fingertip-worn haptic interface is useful and shows practical engineering effort.
4. The use of consumer-grade hardware strengthens the practical value of the prototype and supports its potential for accessible training environments.
5. The technical validation against XPBD and VegaFEM provides useful evidence of real-time performance and deformation stability.
6. The formative expert evaluation, although limited, gives meaningful feedback on system feasibility, force perception, and remaining clinical realism issues.
7. The authors are transparent about several limitations, including simplified anatomical constraints, lack of nonlinear tissue behavior, and limited 1-DOF haptic feedback.

Major Issues

The main contribution needs to be clarified more strongly. The paper presents the work as a system-level integration of GB-cFEM, hand tracking, and a fingertip haptic device, but it is not fully clear what is new compared with the authors' previous GB-cFEM and haptic-device studies.

1. The evaluation is too limited to support the broader claims made in the paper. Only one surgical expert participated, so the findings should be treated as early formative feedback rather than evidence of training effectiveness or clinical usefulness.
2. The paper discusses medical education and training value, but no actual learning outcomes are measured. There is no evaluation of novice performance, skill

improvement, retention, or transfer to real clinical tasks. These claims should either be reduced or supported with additional experiments.

3. The haptic feedback system needs stronger quantitative validation. The manuscript reports the force-mapping equation and update rate, but it does not provide enough evidence about actual force accuracy, latency, calibration, actuator limits, force resolution, or stability during interaction.
4. Since lesion localization is the central task, the paper should include objective task-performance results. Metrics such as localization accuracy, localization time, localization error, number of pressing attempts, and comparison between haptics-on and haptics-off conditions would make the evaluation much stronger.
5. The biomechanical realism of the liver model remains a concern. The expert feedback indicates that the simulated tissue felt overly elastic and closer to soft gel than real liver tissue. The selected Young's modulus values, lesion stiffness ratio, lesion size, and boundary constraints need stronger justification from literature, phantom data, or clinical calibration.
6. The comparison with XPBD and VegaFEM is useful, but the benchmark setup is not described in enough detail. The authors should provide clearer information about solver settings, mesh scale, loading conditions, boundary constraints, time steps, substeps, and convergence criteria so that the results are reproducible.
7. The 1-DOF haptic device is a clear limitation for realistic palpation. It can provide normal resistance, but it cannot reproduce tangential force, squeezing sensation, or complex tissue-contact feedback, which are important for realistic soft-tissue interaction.
8. Some conclusions are stronger than the evidence supports. The current study demonstrates technical feasibility and early expert feedback, but it does not yet prove that the system is ready as a clinical-grade trainer or that it improves medical learning outcomes.

Minor Issues

1. Some manuscript formatting details need correction, including placeholder publication information and template-related inconsistencies.
2. The terms "lesion," "tumor," and "hard region" are used somewhat interchangeably. These terms should be defined clearly and used consistently throughout the paper.
3. Some figures and tables should be improved for readability. In particular, the performance plots and system screenshots would benefit from larger labels, clearer captions, and more explanation of experimental conditions.
4. The Likert-scale results should be interpreted carefully because they come from only one participant. The authors should present them as qualitative formative feedback rather than generalizable usability evidence.
5. The supplementary video is useful, but the manuscript should explain more clearly what the video demonstrates, including whether the lesion is visible only for demonstration or also during evaluation.
6. Some recent references, especially 2025 works, appear incomplete or need full bibliographic information. The authors should verify all reference details before resubmission.

7. The manuscript is generally understandable, but some sections are wordy and repetitive, especially in the Introduction and Discussion. A careful language edit would improve clarity and flow.